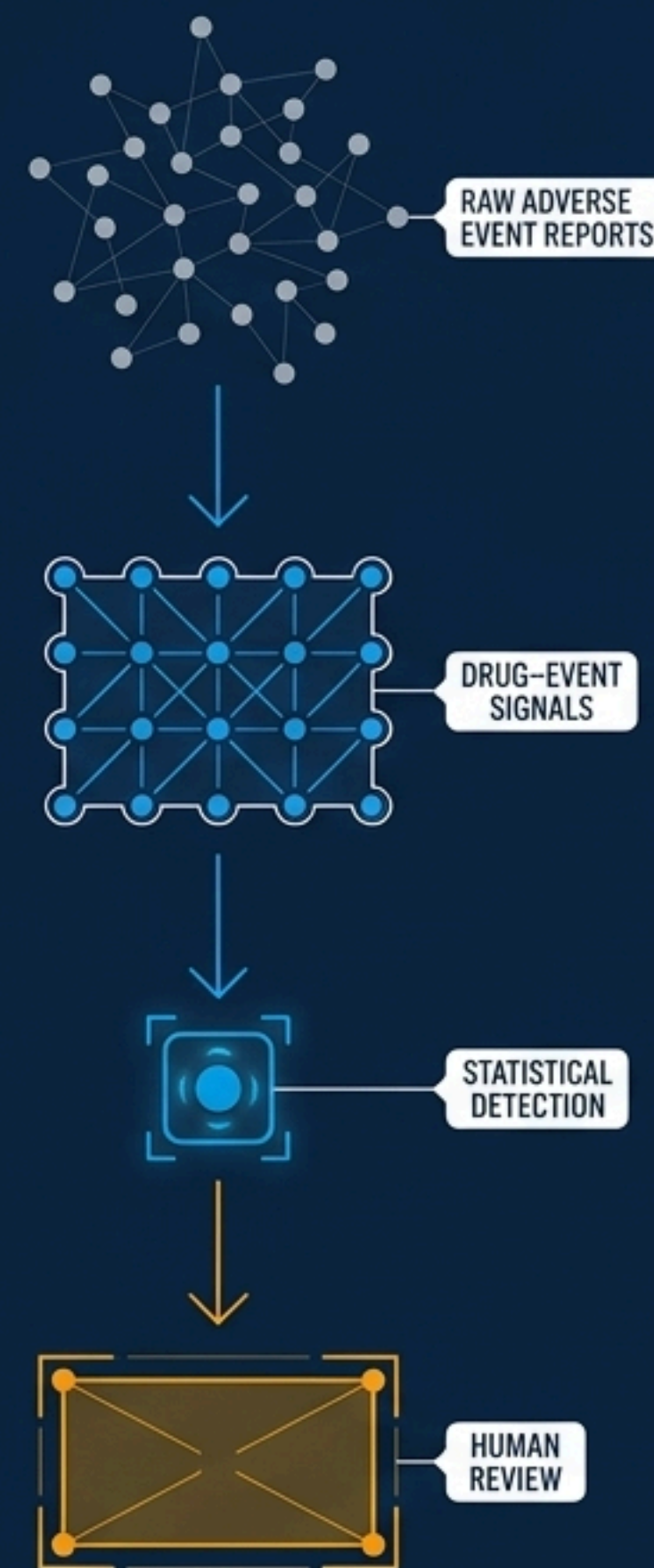


SIGNALTRACE

Intelligent Pharmacovigilance Workspace

Turning adverse-event data into prioritized safety signals.

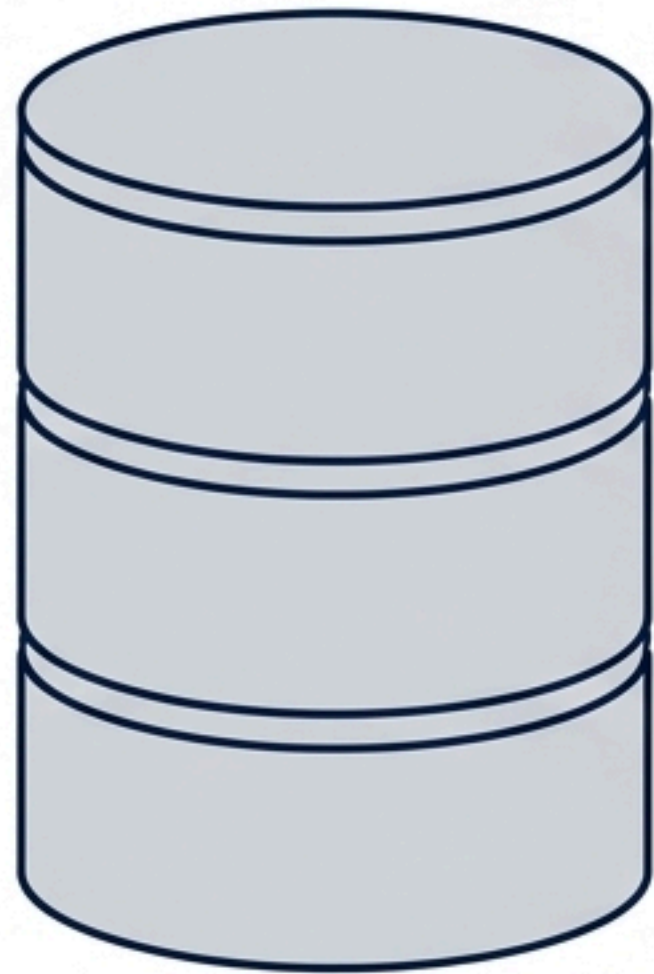
Team VisionX



Safety signals are buried in massive volumes of reports.

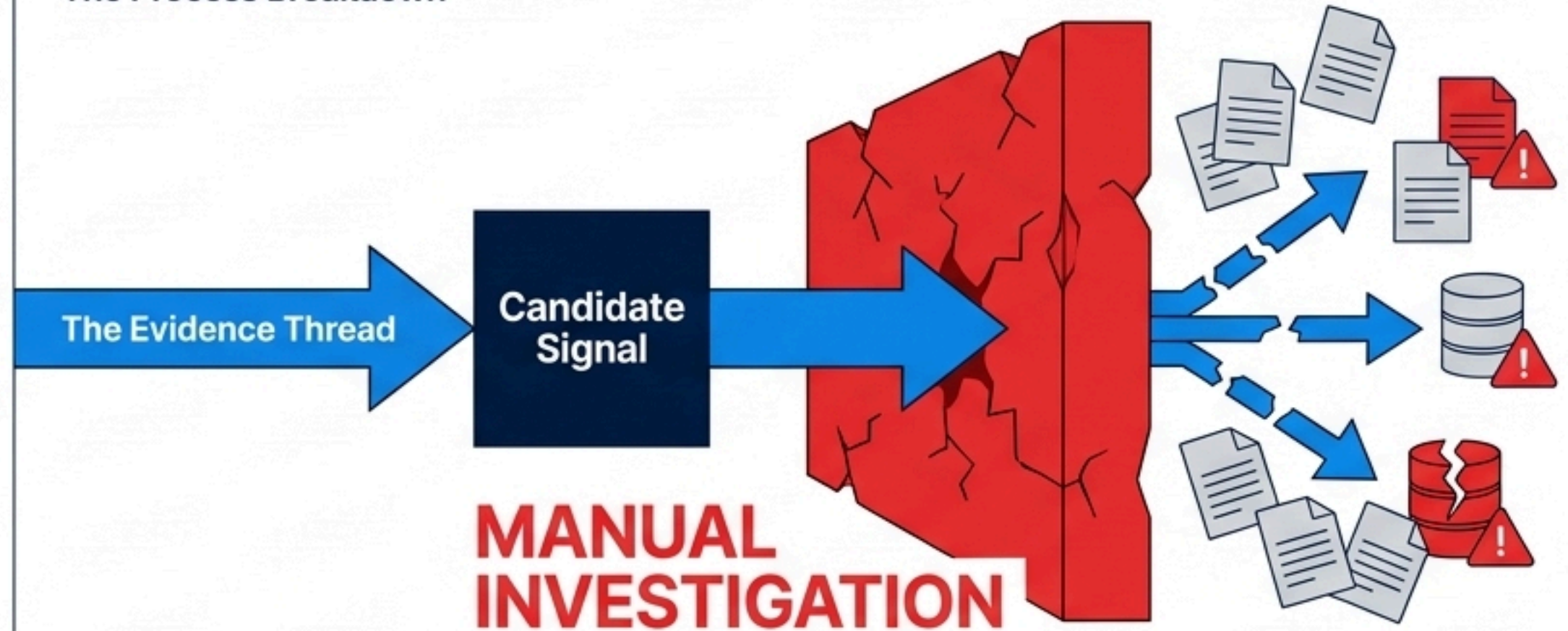
The Macro Truth

Millions of adverse-event reports create an enormous review burden.



**20M+
FAERS
adverse-
event
reports**

The Process Breakdown



01 — DATA VOLUME

20M+ FAERS reports are impossible to manually screen comprehensively.

02 — SIGNAL DISCOVERY

Important statistical patterns remain hidden inside the reporting background.

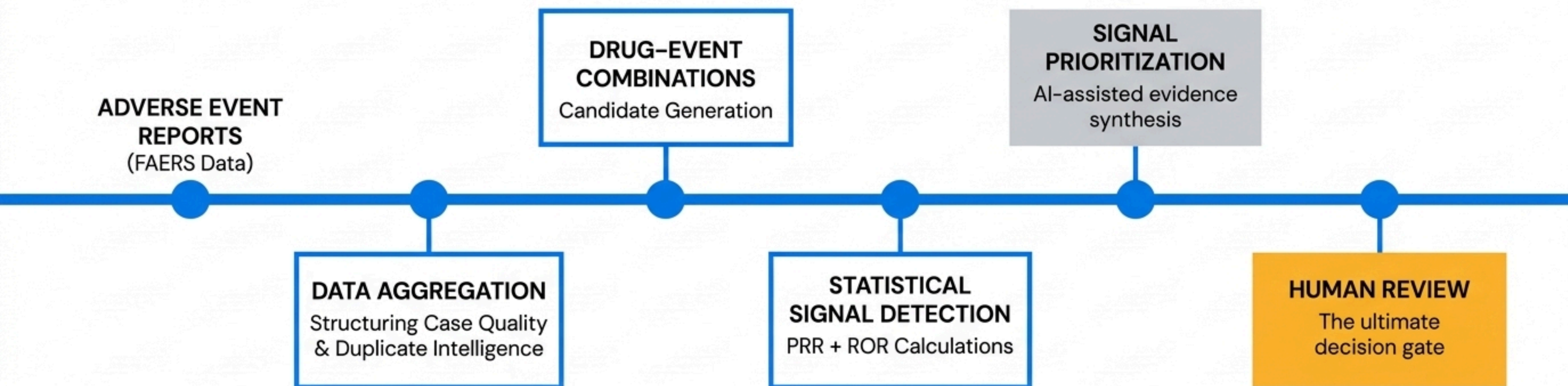
03 — REVIEW PRIORITIZATION

Safety teams lack a structured way to identify which candidate signals deserve closer investigation.

The bottleneck is not finding data. It is turning data into review-ready evidence.

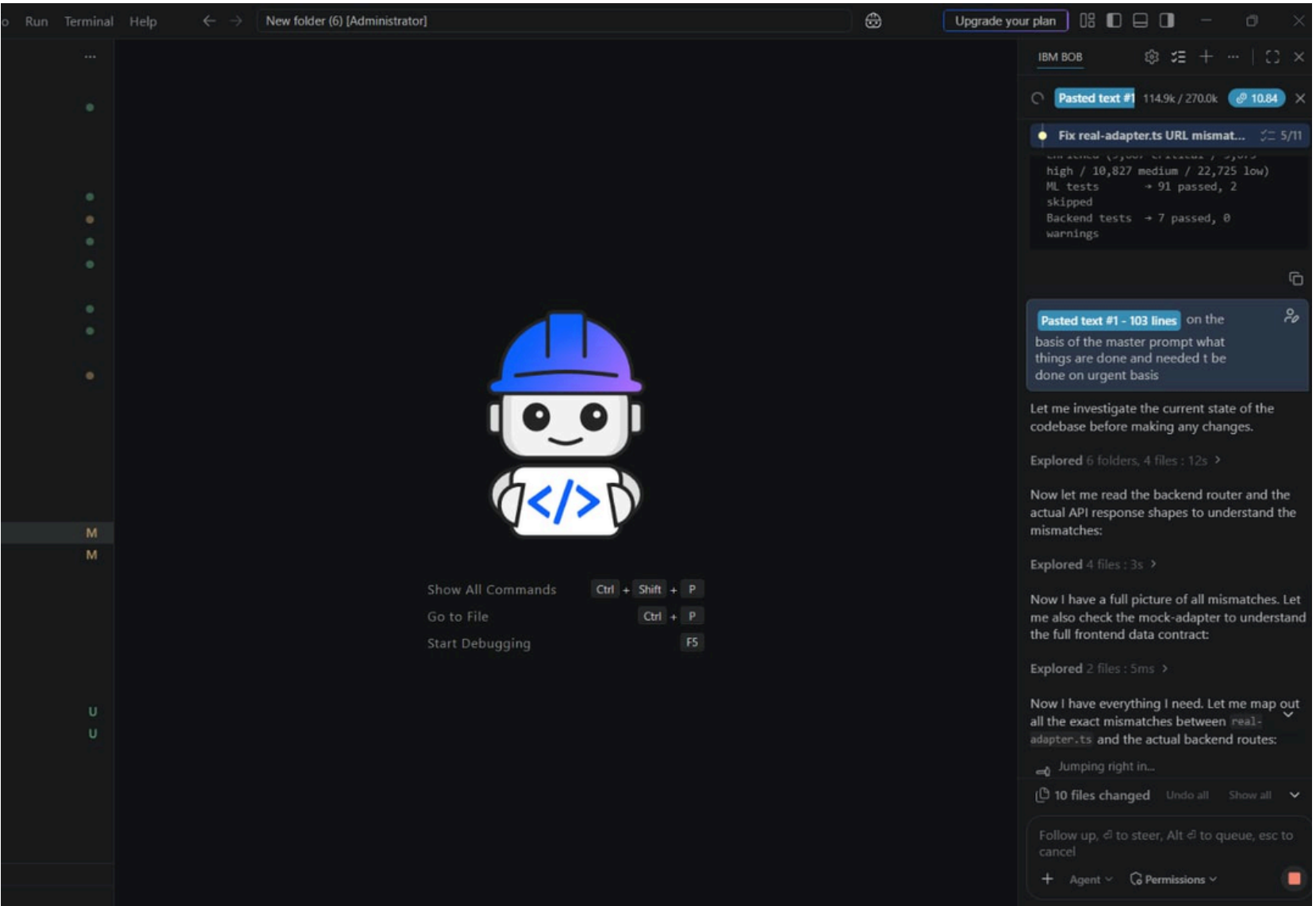
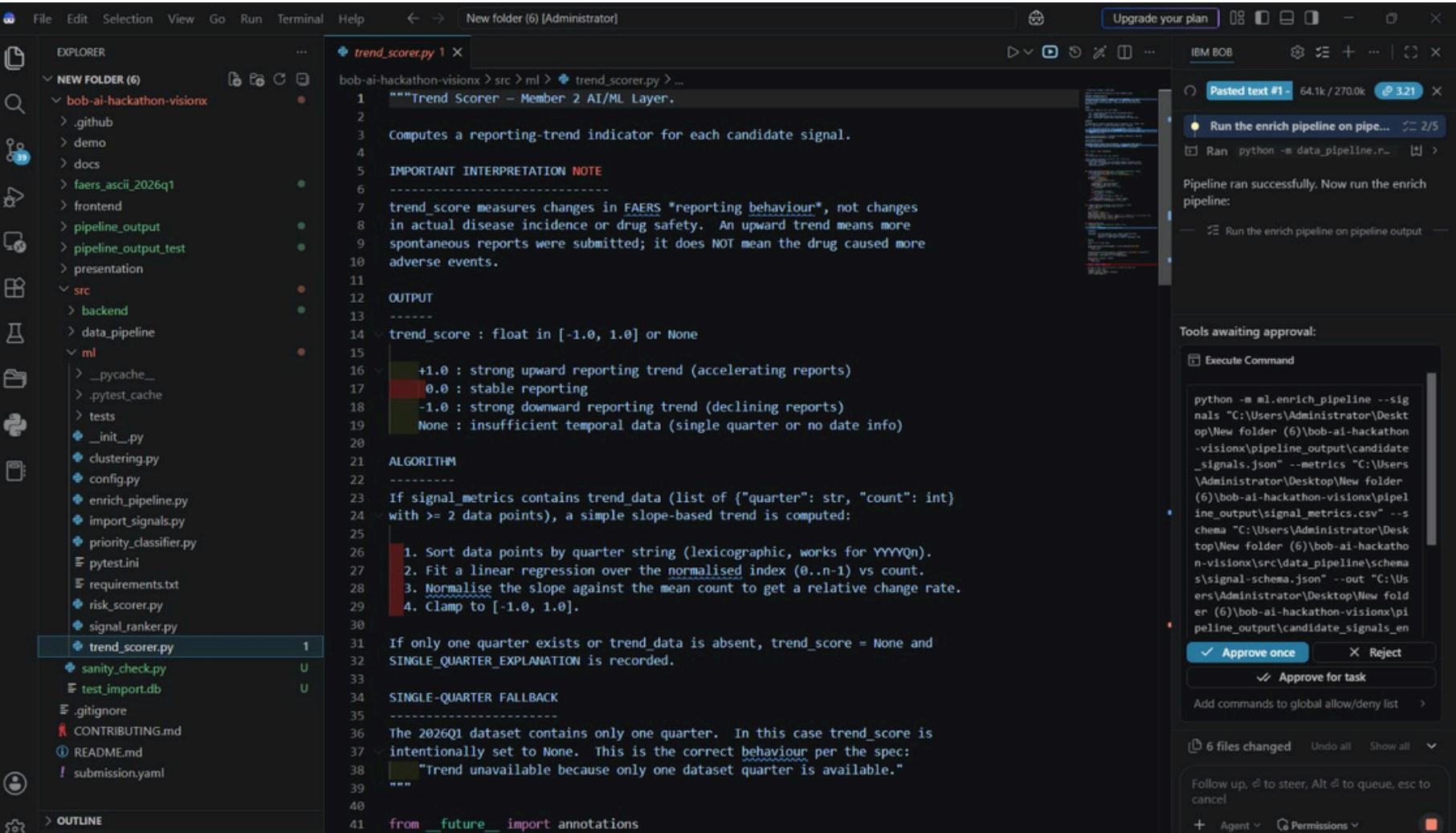
SignalTrace turns reporting data into a review queue.

An AI-assisted pharmacovigilance decision-support platform that analyzes drug-event reporting patterns and prioritizes candidate safety signals for human review.

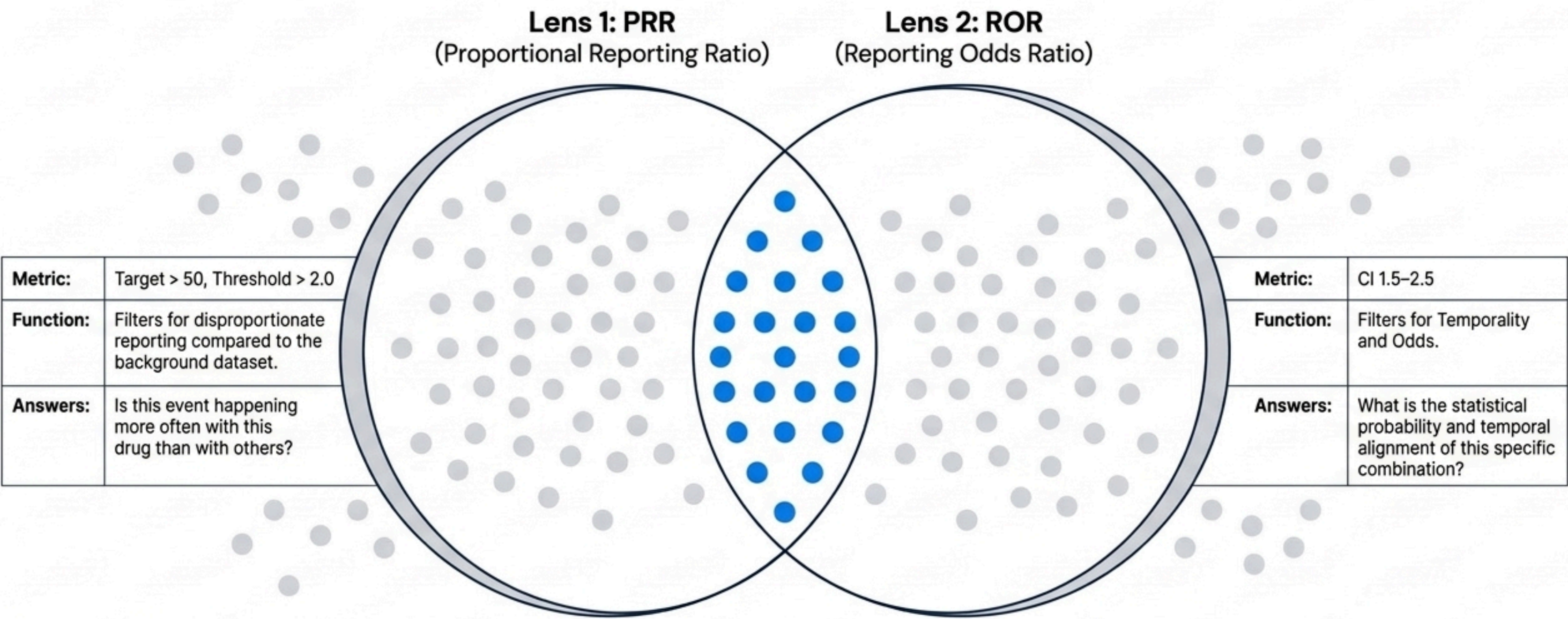


Myth: AI replaces pharmacovigilance experts.
Reality: SignalTrace does not replace experts.
SignalTrace helps experts decide where to look first.

Usage of IBM Bob



Two statistical lenses help identify disproportionate reporting.



Architected for absolute traceability and grounded synthesis.

The Dual-Engine Diagnostic Matrix

Calculated Context (Deterministic Engine)

Core Tech: PostgreSQL / Exact Math.

Role: Owns signal detection, duplicate clustering, and exact PRR/ROR calculations.

Rule: Cannot hallucinate. Provides the immutable statistical facts derived directly from FAERS.

AI Synthesis (Generative Engine)

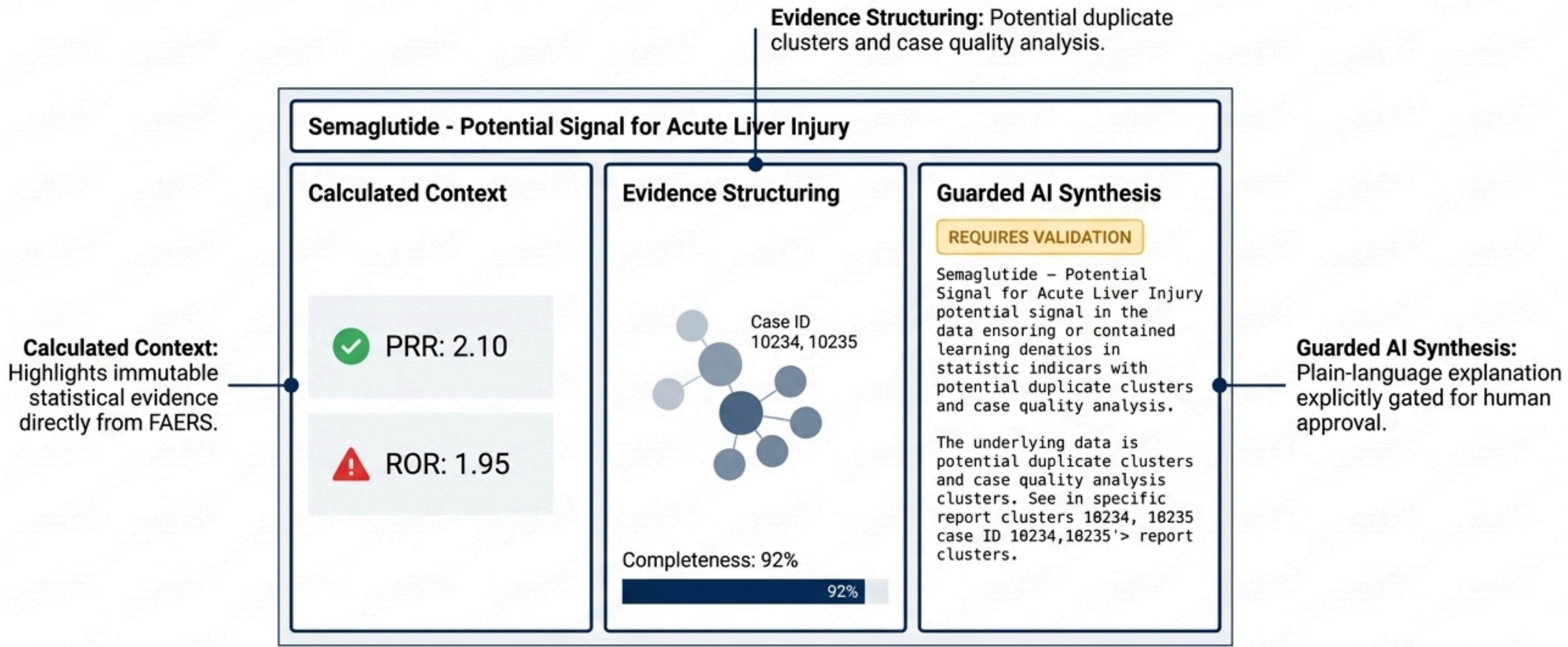
Core Tech: Groq (LLM) / Gemini.

Role: Generates plain-language explanations and draft summaries.

Rule: Strictly Grounded. Restricted solely to summarizing the pre-computed facts handed to it by the Deterministic Engine.

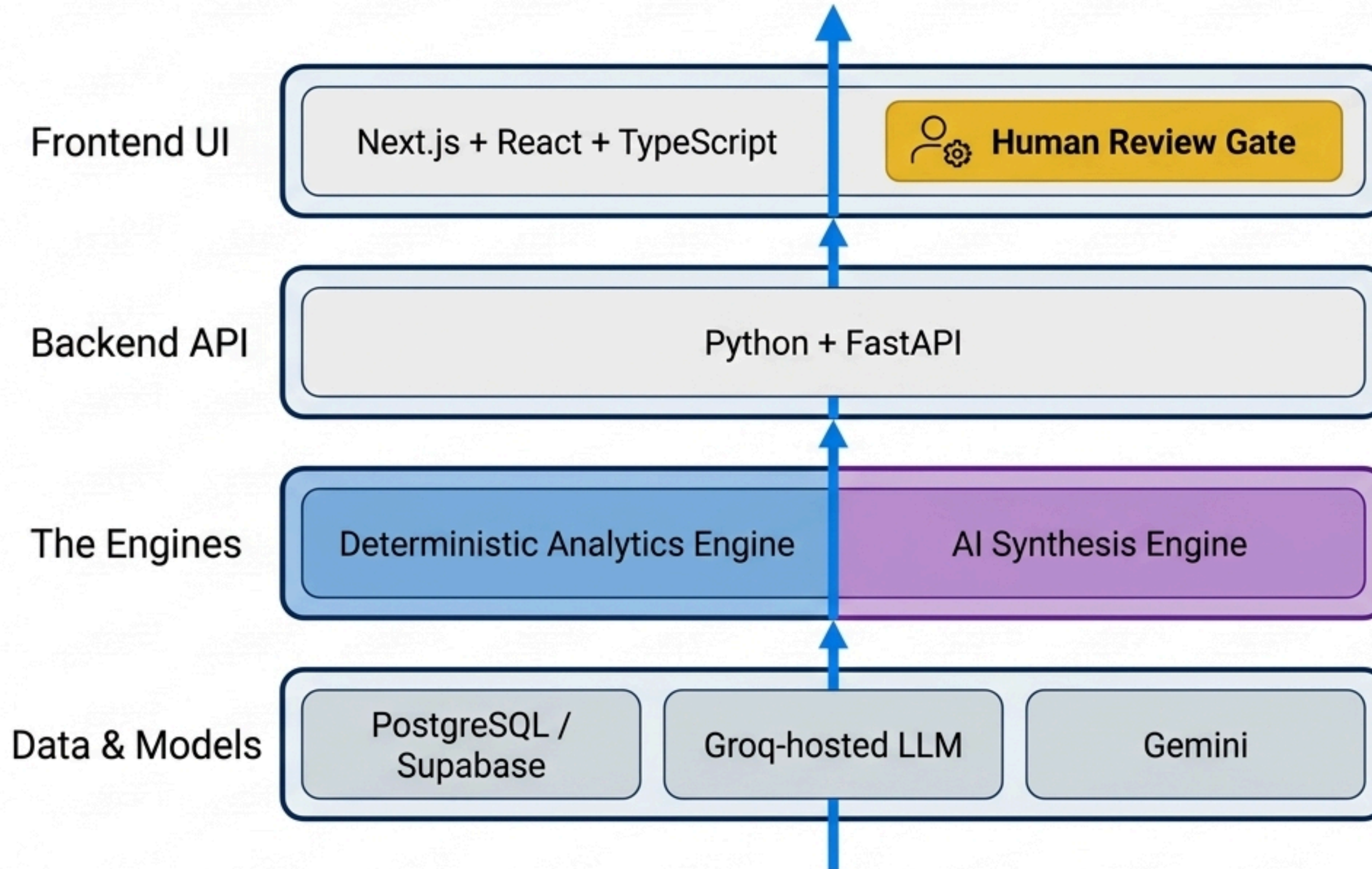
LLMs synthesize the narrative. Deterministic analytics preserve the facts.

A clinical command center built around human validation.



Every AI output requires a human-in-the-loop decision gate.
The interface forces deliberate clinical judgment over passive consumption.

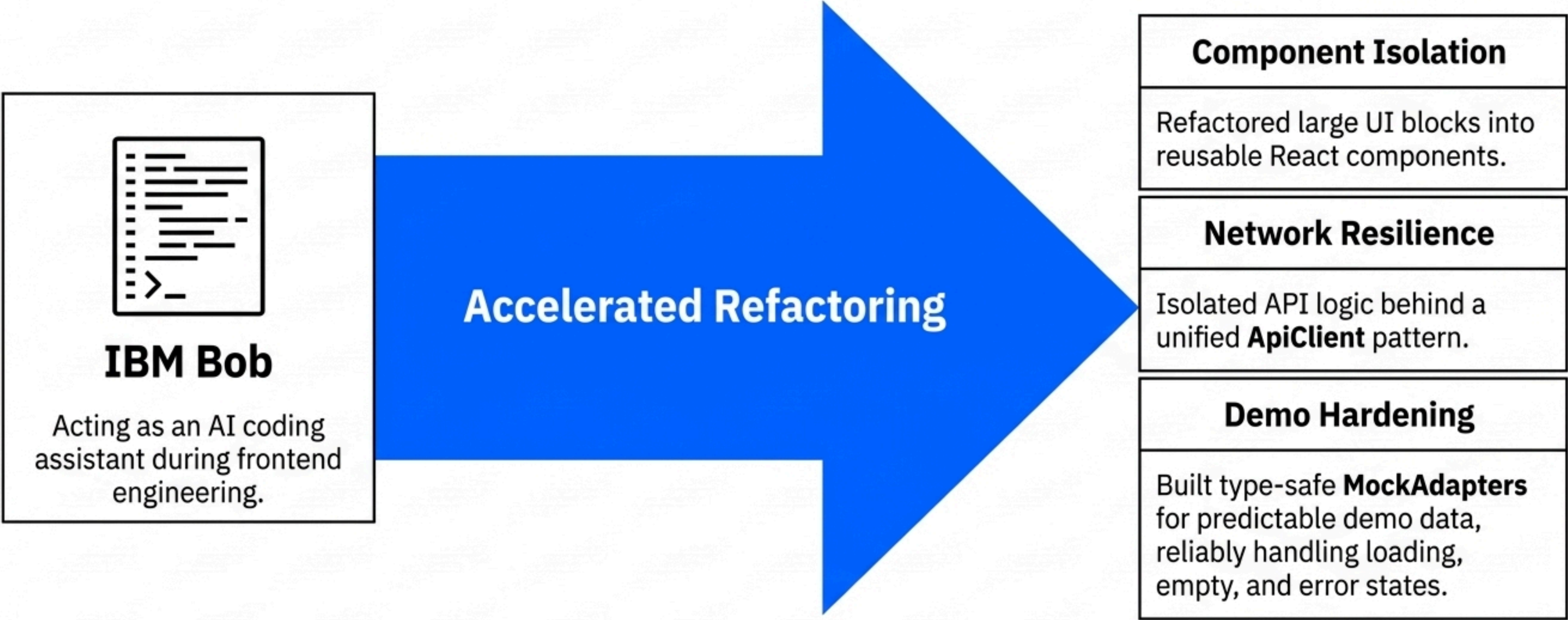
Enterprise-grade separation of concerns.



Traceability Core

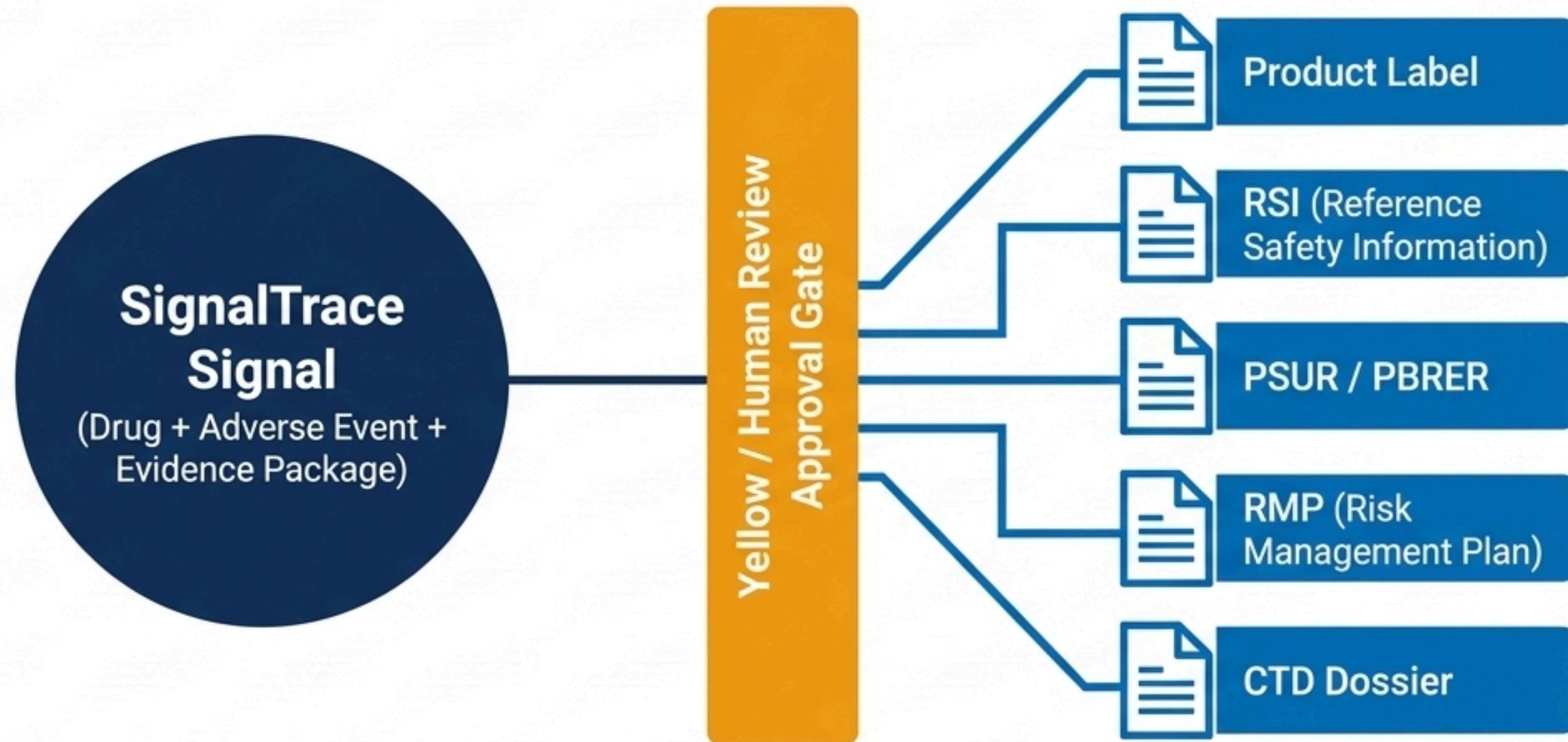
Every statistical claim visible in the Next.js frontend can be traced back through the Python API directly to a specific row in the Supabase PostgreSQL database.

Accelerated engineering via IBM Bob.



Outcome: Faster iteration, a cleaner architecture, and a highly resilient demo experience.

The Signal-to-Regulatory Bridge



Faster Review

Eliminates repetitive, manual evidence-compilation work.

Traceability

Preserves a clear, auditable path from raw signal to document.

Human Expertise

Experts spend time on clinical judgment, not data entry.

The Builders

HARSHIL

THAKKAR

Data Engineering
& Feature
Pipeline

SHARNAM

KATWALA

AI / ML & Signal
Intelligence

MANN SHAH

Backend & NLP /
Regulatory
Intelligence

JIYA SADARIA

Frontend,
UI/UX

Project: SignalTrace

Track: IBM Bob AI Hackathon / Pharmacovigilance & Regulatory Intelligence

Team: VisionX

“SignalTrace turns a candidate safety signal into a traceable, review-ready evidence trail — while keeping humans in control.”